

Zehua Wang

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EDUCATION

- **Massachusetts Institute of Technology — GPA: 5.00/5.00** Cambridge, MA
B.S. Candidate in Physics and Artificial Intelligence and Decision Making Expected May 2028
- **Tsinghua University — GPA: 3.95/4.00** Beijing, China
Preparatory Program and First-Year Coursework, Institute for Interdisciplinary Information Sciences (IIIS) Aug 2024 – Jul 2025

HONORS & AWARDS

- **Gold Medal (1st in Theory)**, 54th International Physics Olympiad (IPhO) Jul 2024
- **1st Place**, MIT Informatics Tournament (MITIT) 2025 Winter Contest Beginner's Round Dec 2025
- **2nd Place, Modal Prize**, HackMIT 2025 Sep 2025

EXPERIENCE

- **Sophomore Intern, Algorithm Development & Software Engineering** New York, NY
Hudson River Trading May 2026 – Aug 2026
 - Developed and live-tested a trading strategy during the Algorithm Development rotation.
 - Contributed enhancements to internal developer tooling in HRT's codebase during the Software Engineering rotation.
- **Undergraduate Researcher, Diffusion Models & Molecular Dynamics** Cambridge, MA
MIT — Advisor: Tommi Jaakkola Feb 2026 – May 2026
 - Co-developed Diffusion Accelerants, a non-Markovian biasing method that augments molecular dynamics with learned measure transport.
- **Undergraduate Researcher, Reinforcement Learning & Embodied AI** Cambridge, MA
FortyFive Labs — Advisor: Ge Yang Oct 2025 – Apr 2026
 - Developed a whole-body control policy for functional compliance, combining explicit force estimation with directional, time-varying stiffness profiles at deployment.
 - Ran parallel RL and imitation-learning experiments across nearly 100 NVIDIA RTX PRO 6000 GPUs; performed extensive sim-to-real deployment, debugging, and policy iteration on Unitree G1 hardware.
 - Contributed code to **Vuer**, a 3D visualization toolkit, and **ML-Dash**, an experiment-tracking and data platform.
- **Undergraduate Researcher, Learning-Based Control** Beijing, China
Tsinghua University — Advisor: Huazhe Xu Mar 2025 – Aug 2025
 - Developed a multistage RIR (RL-to-imitation-to-real-world) pipeline for Franka manipulation, combining PPO/DrQ-v2 specialists, multitask imitation learning, and sim-to-real transfer.

PUBLICATIONS

- **Diffusion Accelerants: Towards Augmenting Molecular Dynamics with Learned Measure Transport**
Bowen Jing, **Zehua Wang**, Paul Gutkovich, Peter Holderrieth, Tommi Jaakkola.
SPIGM Workshop at ICML 2026 (poster). [Paper]
- **Family-Aware Residual Architecture for Predicting Quantum Circuit Simulation Performance**
Honjar Xing*, Yehong Jiang*, Xianbang Wang*, **Zehua Wang***, Zhicheng Jiang*.
QC-CSAA Workshop at IEEE ISVLSI 2026 (full paper). [Paper] * Equal contribution.

SELECTED PROJECTS

- **Open-Source Contributor, NousResearch/Hermes Agent**: Authored a **3-commit** speech-to-text provider-validation patch now in **main**, adding selected-provider and plugin-backend checks plus tests. [Commits] [Review]
- **Video Real2Sim (VR2S) (MIT 6.8300)**: Co-built a single-image real-to-sim pipeline that synthesizes orbital views with a video model and reconstructs them with a pose-free 3D stack; improved perceptual fidelity and back-side appearance while characterizing a silhouette-accuracy tradeoff. [Project]
- **Fast Humanoid Loco-Manipulation via Flow Matching (MIT 6.4210)**: Compared DDPM and Flow Matching for humanoid trajectory synthesis; at five function evaluations, Flow Matching achieved **820 vs. 280 survival steps**. Enabled zero-shot loco-manipulation from walking-only data via test-time classifier guidance. [Code] [Video]
- **RL vs. SFT for Mathematical Reasoning in LLMs (MIT 6.4610)**: Implemented GMPO and its evaluation pipeline for a compute-controlled comparison of PPO, GRPO, GMPO, and RLOO with SFT on Qwen3-8B; GMPO reached **74.2%** on GSM8K versus 76.7% for SFT. [Project] [Code]

SKILLS

Programming: Python, C++ — **ML/Robotics:** PyTorch, MuJoCo, Isaac Gym, Isaac Lab — **Systems:** Slurm, Docker, Linux